

Pranoy Ray

Postdoctoral Researcher @ Argonne National Laboratory & University of Chicago

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Summary

Computational materials researcher and data scientist specializing in molecular structure screening using spatial statistics.

Expertise in feature engineering workflows that inform robust material process-structure-property relationships across chemical processes.

Experienced integrating physics-based constraints with data-driven models on HPC systems, with a track record delivering R&D for chemical, semiconductor, and federal lab partners.

Education

PhD	Georgia Institute of Technology , Mechanical Engineering	Atlanta, GA, USA
	• Advisor: Dr. Surya R. Kalidindi	Aug 2021 – May 2026
	• Thesis: On efficient screening strategies for molecular structures	
MS	Georgia Institute of Technology , Computational Science and Engineering	Atlanta, GA, USA
		Aug 2022 – May 2025
BTech	National Institute of Technology, Durgapur , Metallurgical & Materials Engineering	Durgapur, India
		July 2016 – July 2020

Experience

Multiscale Technologies Inc. , Data Science Manager & Materials Scientist Intern	Seattle, WA, USA
• Designed and deployed active learning workflows to accelerate process optimization, minimizing experimental trial-and-error loops for semiconductor, chemical, and US national lab partners.	Jan 2024 – Aug 2024
• Supervised and led a team of data scientists and engineers across the USA, France, India, and Pakistan delivering exploratory materials design solutions for Fortune 100 clients.	8 months
• Architected the integration of physics-based material simulations into the MIND ecosystem, unifying pipelines across engineering APIs and cross-functional teams under MOT/Root Cause Analysis protocols.	
Bhabha Atomic Research Centre, HP & SRPD , Research Assistant (Advisors: Dr. Srikumar Banerjee & Dr. Brahmananda Chakraborty)	Mumbai, India
• Established theoretical design principles for solid-state hydrogen storage using DFT and AIMD; published multiple papers.	May 2019 – Aug 2021
	2 years 4 months
Hindustan Aeronautics Limited, Foundry & Forge Division , Project & Industrial Intern (Advisor: Soumya Mandi)	Bangalore, India
• Conducted root cause analysis on manufacturing defects in investment casting, applying process control measures for yield improvement.	May 2018 – July 2018
• Developed preventive measures for quality control using Lean management.	3 months

Awards

William H. Glenn Sr. Fellow	Atlanta, GA, USA
George W. Woodruff School of Mechanical Engineering, Georgia Tech	2026

Woodruff School Fellow Research Excellence, GWW School of Mechanical Engineering, Georgia Tech	Atlanta, GA, USA 2025
Novelis Graduate Scholar Novelis Innovation Hub & Novelis Inc.; featured in GTRI News for research in sustainability, techno-economics of circularity, and AI/data science in materials and manufacturing.	Kennesaw, GA, USA 2024
ELIF Fellow TokyoTech & Strategic Energy Institute @ GT; invited scholar at the 2nd Energy & Informatics Forum.	Honolulu, HI, USA 2022
CMS3 Fellow NSF + Texas A&M University	College Station, TX, USA 2022

Certifications

Management of Technology (MOT) Scheller College of Business, Georgia Tech	2025
Computational Materials Science & Engineering (CMSE) School of MSE, Georgia Tech	2023

Service

- Reviewer & Committee Member — SciPy Conference (2023, 2024, 2025)
- Peer Reviewer (AI4Mat) — NeurIPS 2025, CVPR 2025, PEARC 2025, ICLR 2026
- Peer Reviewer — Springer Nature, Journal of Materials, Chemical Papers, Artificial Intelligence Review, npj Computational Materials
- Session Chair & Reviewer — GT Undergraduate Research Symposium (2025)
- Board Member — Emerging Leaders Advisory Board @ GT (2025-2026)
- Internal VP — Mechanical Engineering Grad Association (MEGA) @ GT (2023-2024)
- President — Entrepreneurship Development Cell, NIT Durgapur (2016-2020)
- Treasurer — Strokes (Art & Photography Club), NIT Durgapur (2017-2020)

Skills

Machine Learning / AI: Neural Networks, Gaussian Process Regression, Normalizing Flows, Autoencoders, Convolutional Networks

Programming & Data Science: Python (PyTorch, GPyTorch, BOTorch, PyVista, SciPy, scikit-learn), R, Java, C

Computational Materials Science: VASP, LAMMPS, GROMACS, DFT, Molecular Dynamics, CG-Martini3, ORCA

HPC / Cloud Platforms: MPI, AWS, GCP, Azure, Shell Scripting (SLURM/PBS), Flask, Firebase, Hadoop, Big Data

Autonomous Design & Decision Making: Bayesian Optimization, Active Learning, Sequential Decision Making, Pareto Frontier Analysis, Design of Experiments